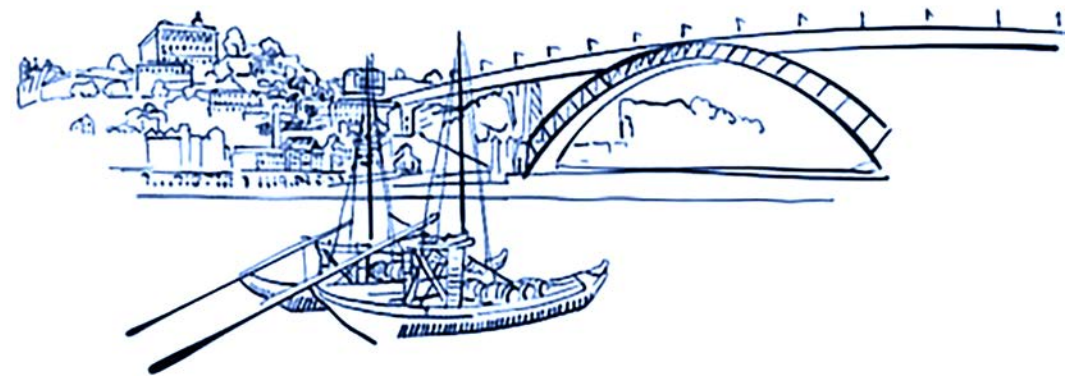




Route-Aware Edge Bundling for Visualizing Origin-Destination Trails in Urban Traffic

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3. University of Groningen



■ Introduction

- OD Trails in Urban Traffic
- Prior Edge Bundling Methods
- Limitations of KDEEB

■ Route-Aware Edge Bundling

- Preprocessing:
 - map matching → hierarchical route structure construction → trail abstraction
- Bundling
 - optimal kernel size setting → density map generation
- Evaluation
 - Bundle termination
 - Bundle deviation

■ Conclusion and Future Work

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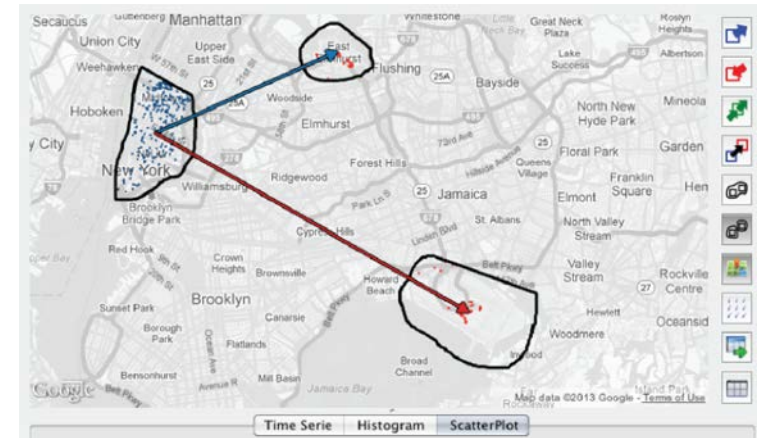
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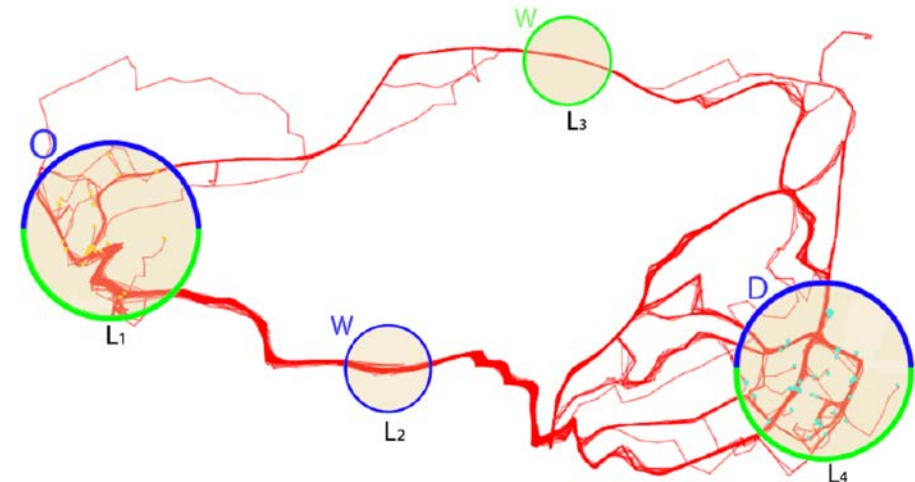
■ Conclusion and Future Work

OD Trails in Urban Traffic

- Urban traffic data, e.g.,
 - Taxi trips in New York, Beijing, Shenzhen
 - Public transportation data in Singapore
 - Electric scooter tracks in Stuttgart
- Origin-destination (OD) is a fundamental concept in transportation, summarizing (people/vehicle/good) movements across geographic locations.
- Properties of OD trails in urban traffic
 - Locations
 - Times
 - Road network
 - Multi-modes



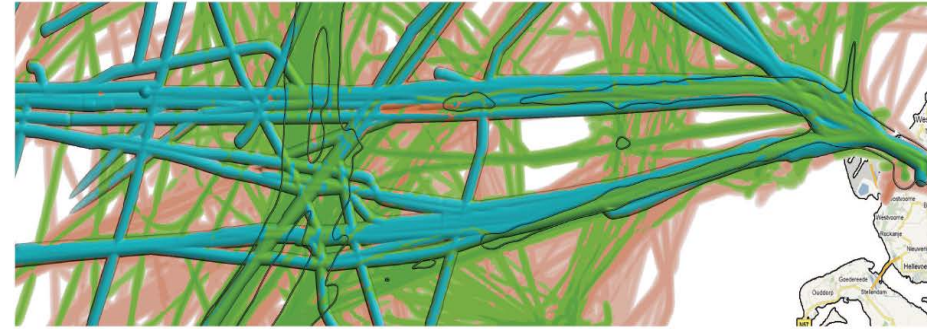
[Ferreira et al. 2013]



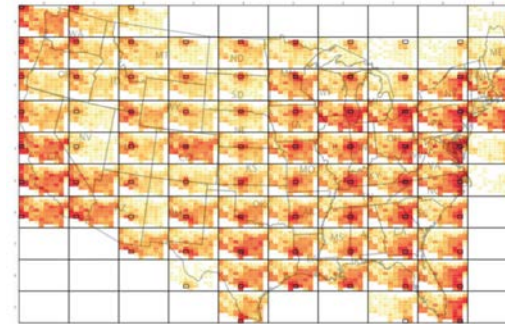
[Krüger et al., 2013]

OD Trail Visualization

- Density Map
 - Summarize trajectories and overview distribution.
- Spatial Aggregation
 - Partition underlying territory into appropriate areas.
- Map Matching
 - Align position records with road network data.
- Direct depiction
 - Directly plot trajectories into 2D/3D displays.



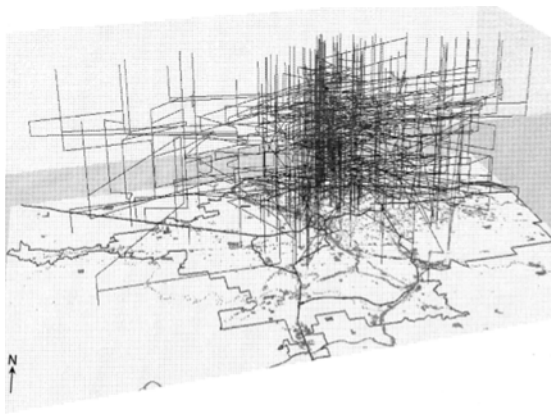
[Scheepens et al., 2011]



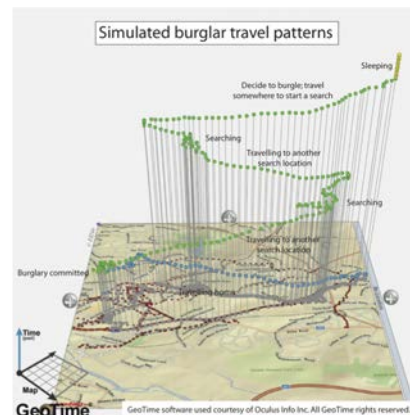
[Wood et al., 2010]



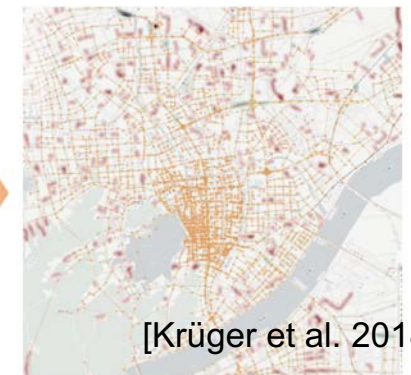
[Andrienko and Andrienko, 2011]



[Kwan, 2000]



[Kapler and Wright, 2004]



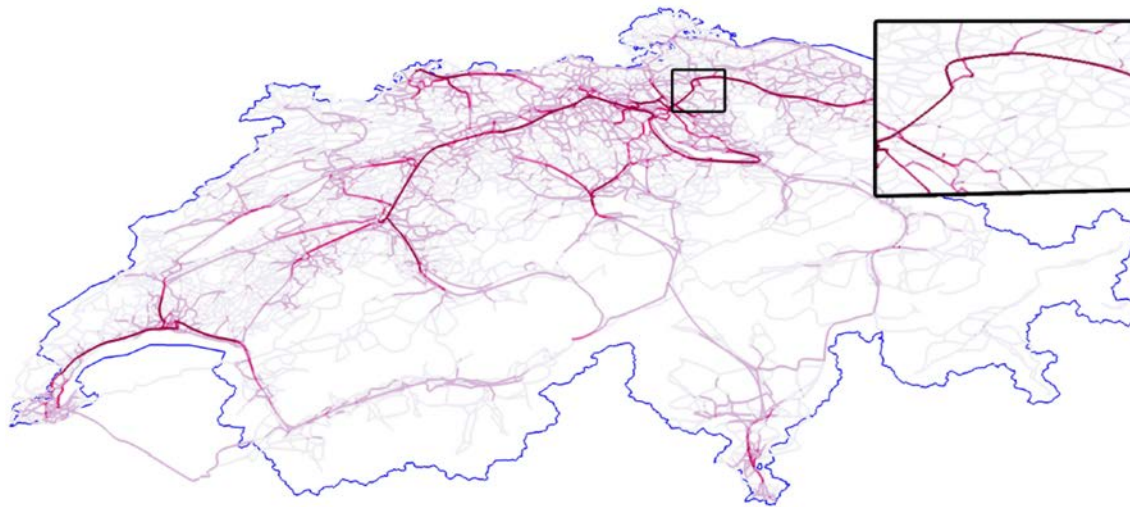
[Krüger et al. 2018]

Prior Edge Bundling Methods

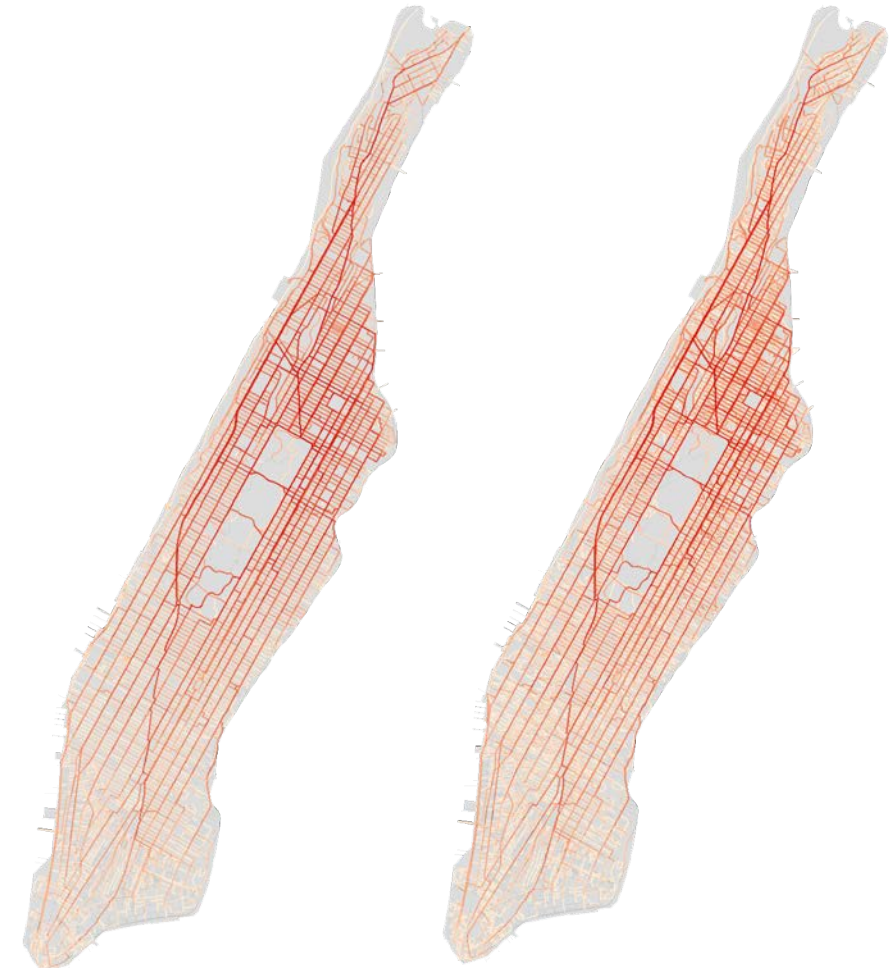
- *Geometry-based* methods: Use control mesh to specify how similar edges are routed.
 - Pros: Flexible to make control mesh
 - Cons: Constructing control mesh can be (very) slow
- *Force-based* methods: Model interaction between spatially close trails as a force field.
 - Pros: No need to make external control mesh
 - Cons: Slow – cannot handle a few thousands trails at interactive rates
- *Image-based* methods: Employ image-processing methods to accelerate the bundling process.
 - Pros: Feasible for GPU implementation – can process millions of trials at interactive rates.
 - Cons: No consideration of spatial constraints when applied to OD trails.

Prior Edge Bundling Methods

- *Constrained Bundling*: Specific constraints are considered.
 - Ambiguity
 - 3D curved surfaces
 - Directions
 - Obstacles avoidance
 - **Vector map**



Vector map for Swiss commuter data
[Thöny & Pajarola, 2015]



Map matching

Vector map

Kernel Density Estimation Edge Bundling (KDEEB)

- We chose KDEEB for the basis of our method:
 - Fast in speed, meanwhile simple enough to implement
 - Be able to incorporate specific constraints

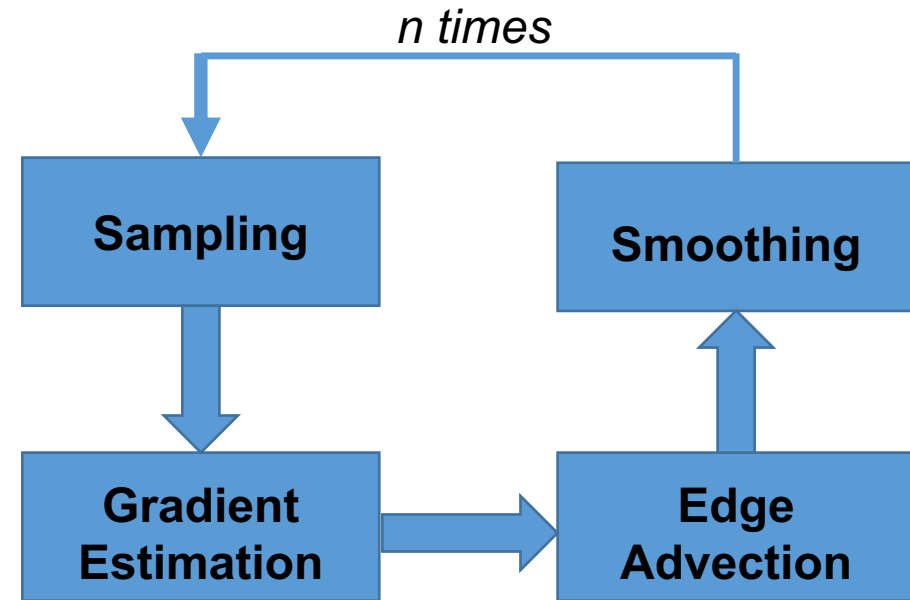
- KDEEB pipeline

- Sampling
- Gradient estimation

$$\rho(\mathbf{x} \in \mathbb{R}^2) = \sum_{\mathbf{y} \in D} K\left(\frac{\|\mathbf{x} - \mathbf{y}\|}{p_r}\right)$$

- Advection
- Smoothing

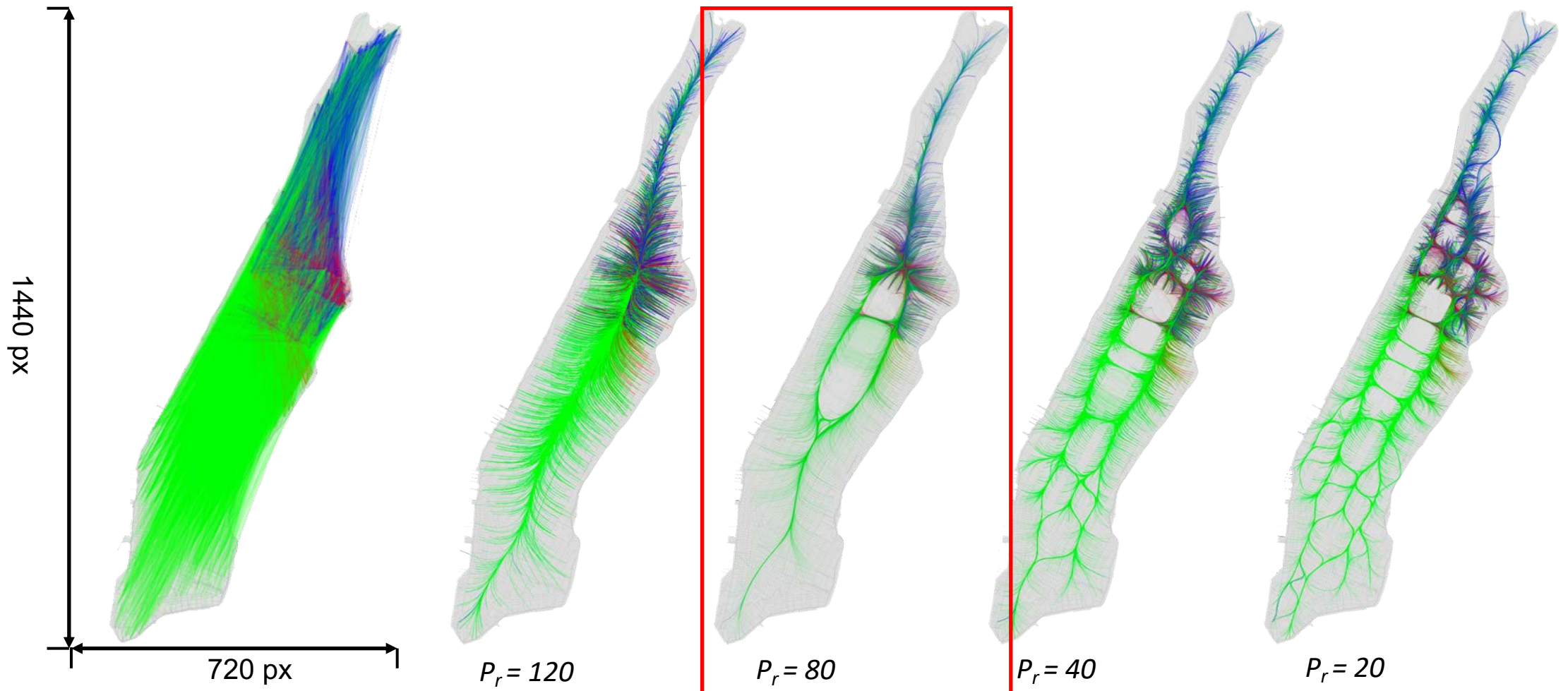
- Iterate n times until stable layout
 - Predefined 10 or 15 times
 - Automatically determined at runtime?



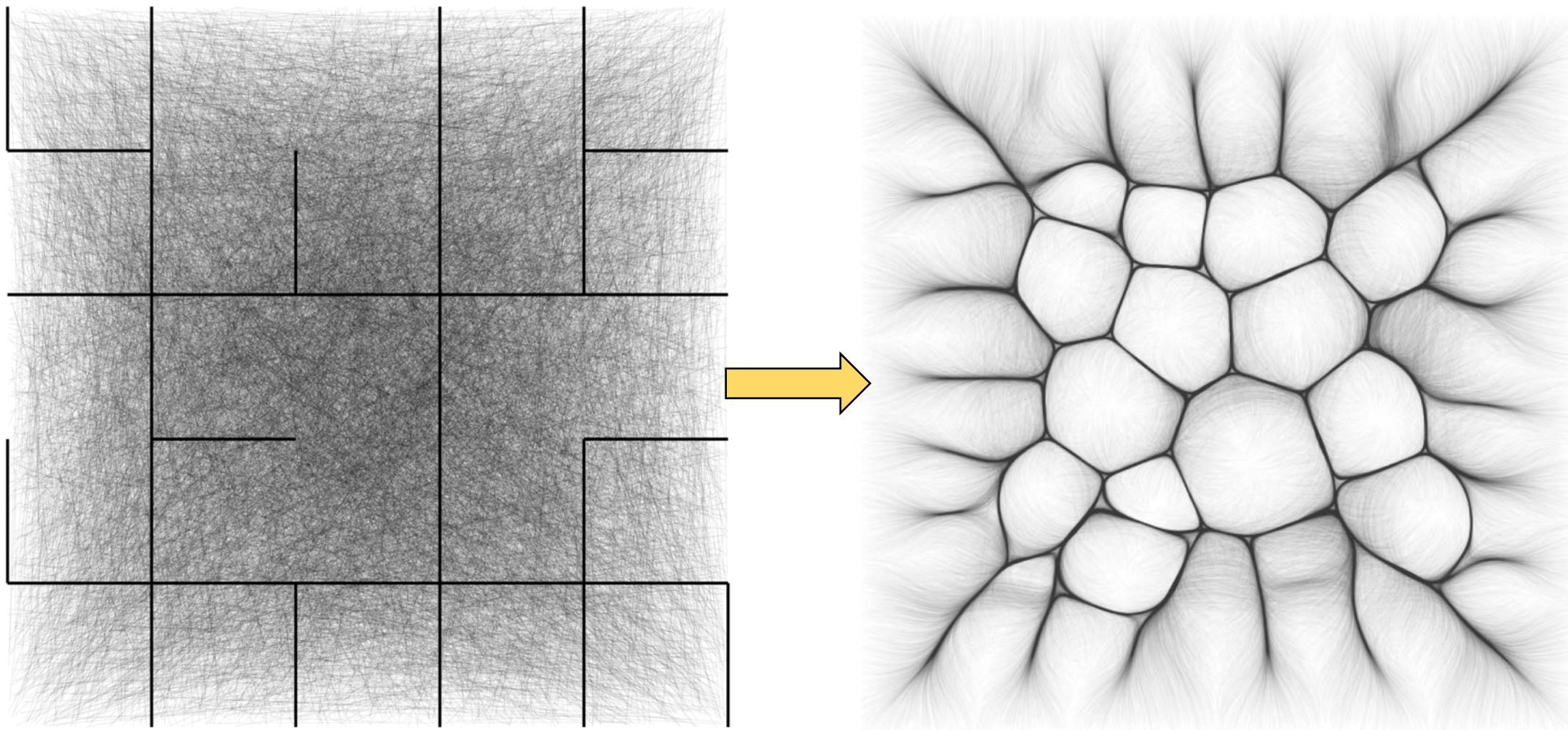
Limitations of KDEEB: What is a suitable p_r ?

- KDEEB: 5% of graph drawing size

- $5\% \times \sqrt{1440^2 + 720^2} = 80.5$



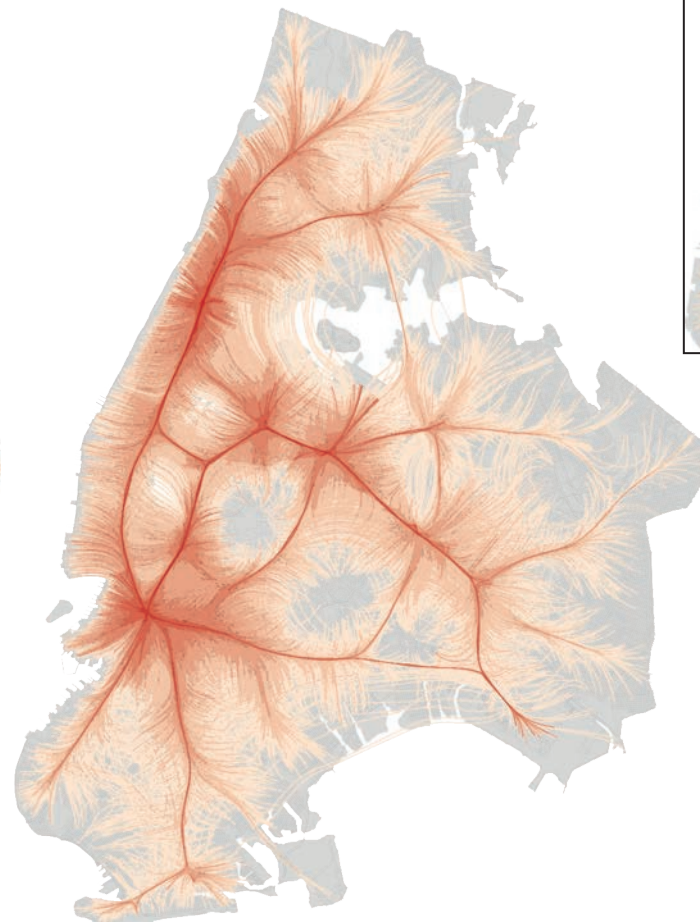
Limitations of KDEEB: Road neglect



Limitations of KDEEB



Map Matching



KDEEB ($p_r = 60$)



Artifacts

KDEEB ($p_r = 21$)

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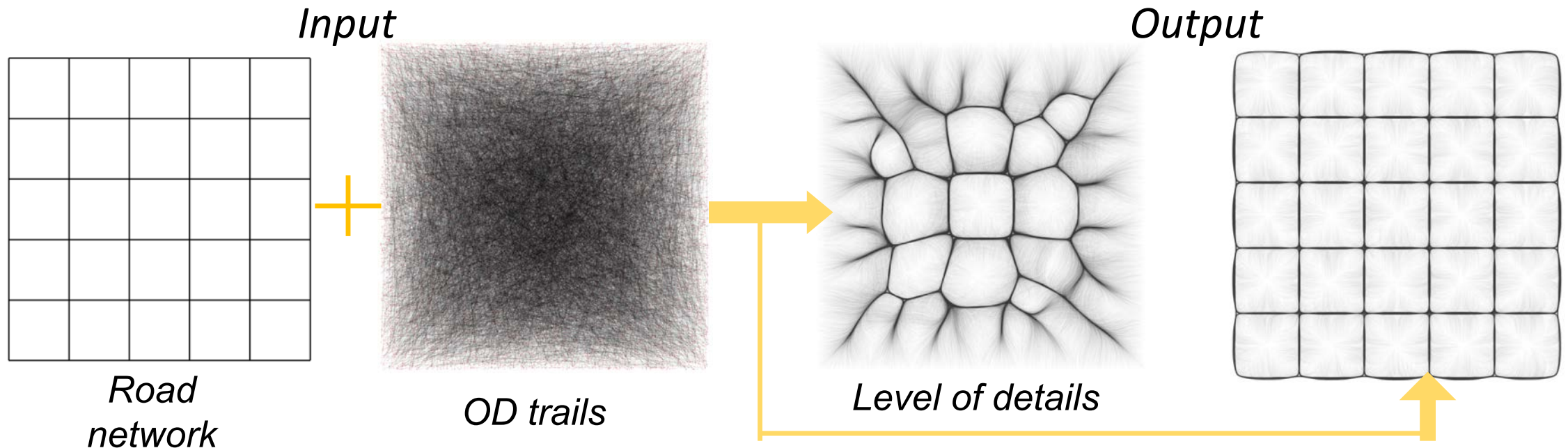
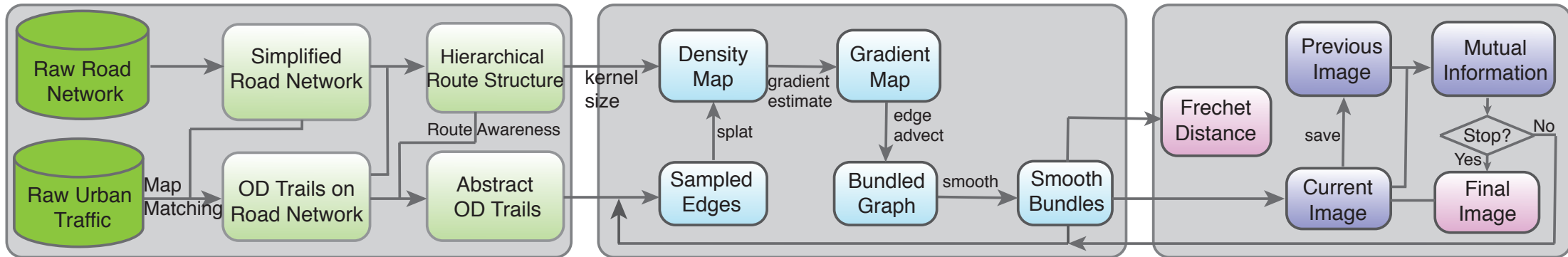
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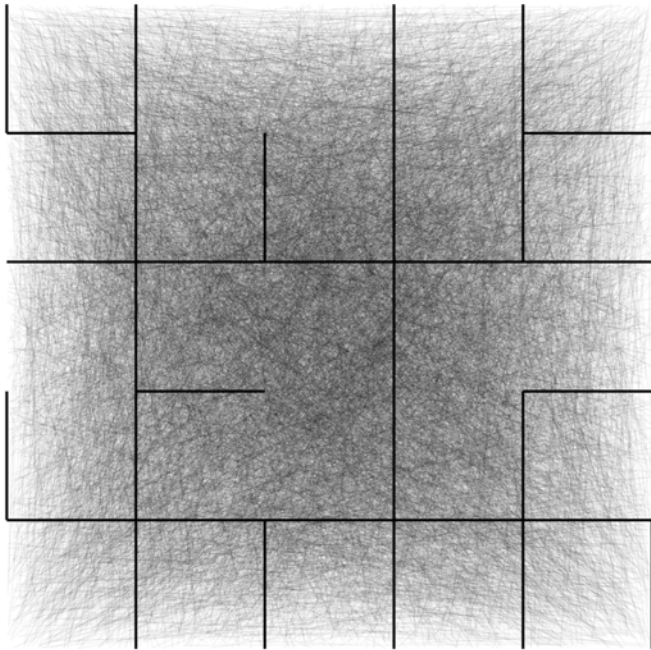
Route-Aware Edge Bundling

- RAEB pipeline: 1) Preprocessing, 2) Bundling, and 3) Evaluation

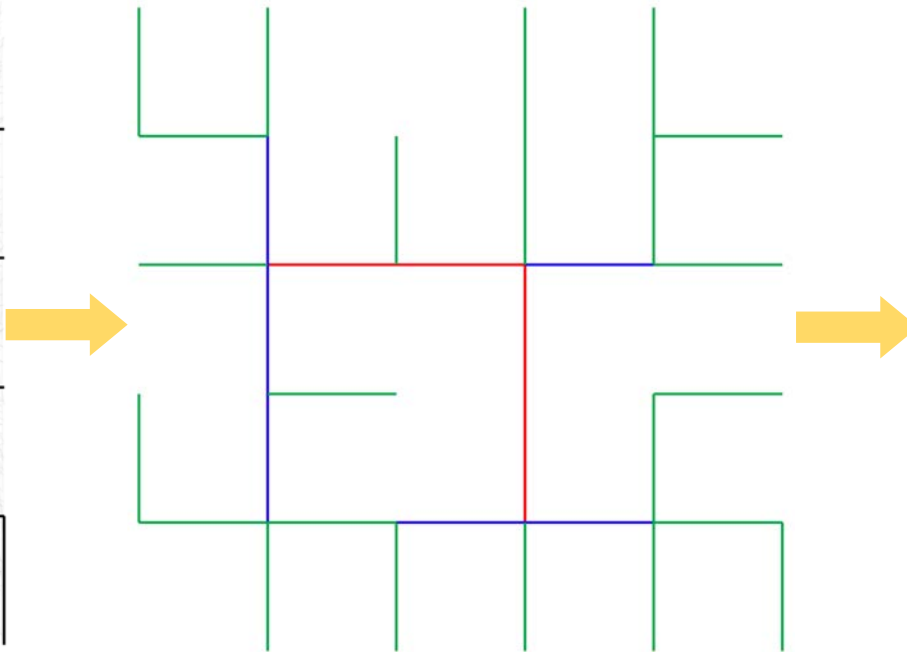


Preprocessing

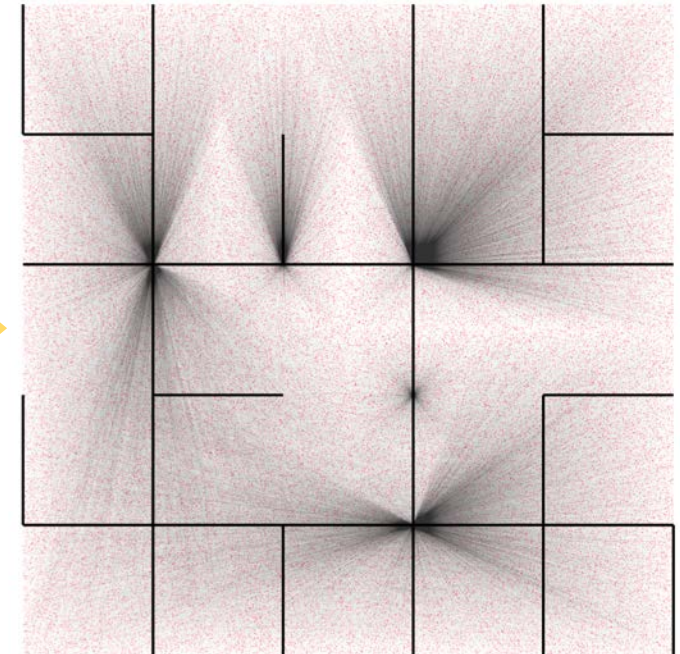
- Build a simplified hierarchical road and traffic network representation.
 - Map matching: shortest path for OD only, ST-matching for GPS traces
 - Hierarchical structure construction: route length, road hierarchy, flow magnitude
 - Trail abstraction: route awareness (p_{ra})



*OD Trails &
Road network*



*Hierarchical
route structure*



Trail Abstraction

Bundling

- KDEEB applied to the hierarchical structure.
 - Optimal kernel size setting
 - Density map generation

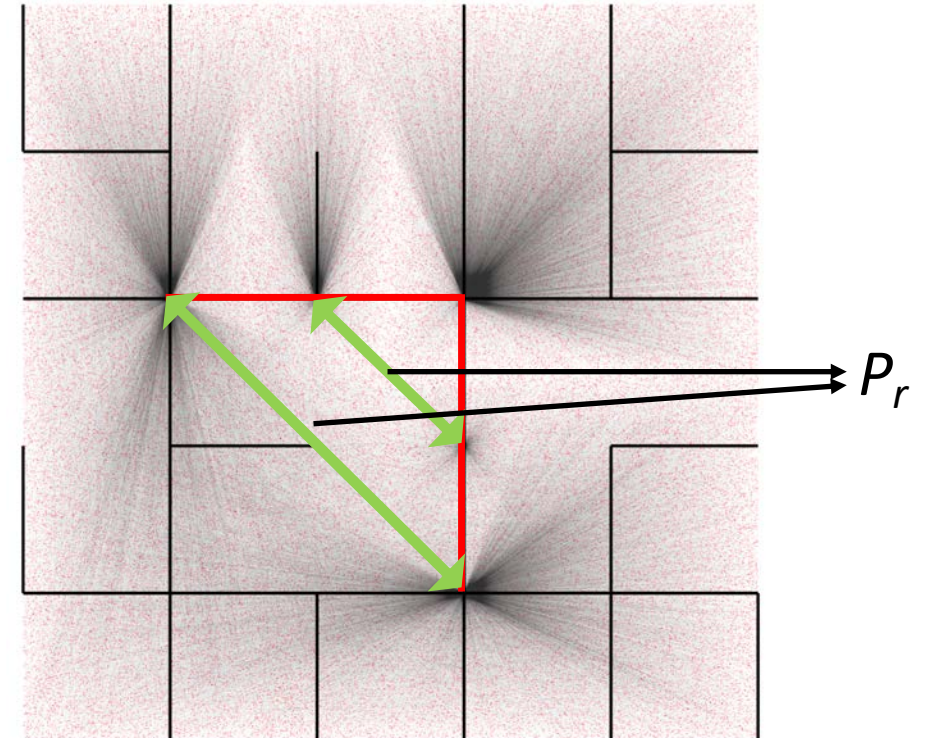
$$\rho_{raeb}(\mathbf{x} \in \mathbb{R}^2) = \sum_{\mathbf{y} \in D} K\left(\frac{\|\mathbf{x} - \mathbf{y}\|}{p_r}\right) + \theta \sum_{\mathbf{r} \in R_{aware}} \Theta(\|\mathbf{x} - \mathbf{r}\|),$$

Algorithm 1 KernelSizeSetting

Input: Top N routes $P = \{P_1, \dots, P_N\}$

Output: Initial kernel size p_r

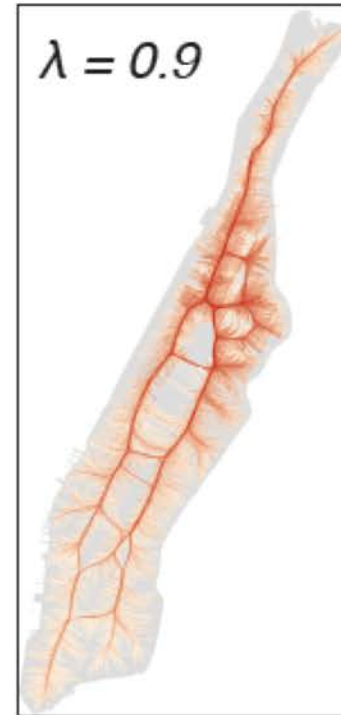
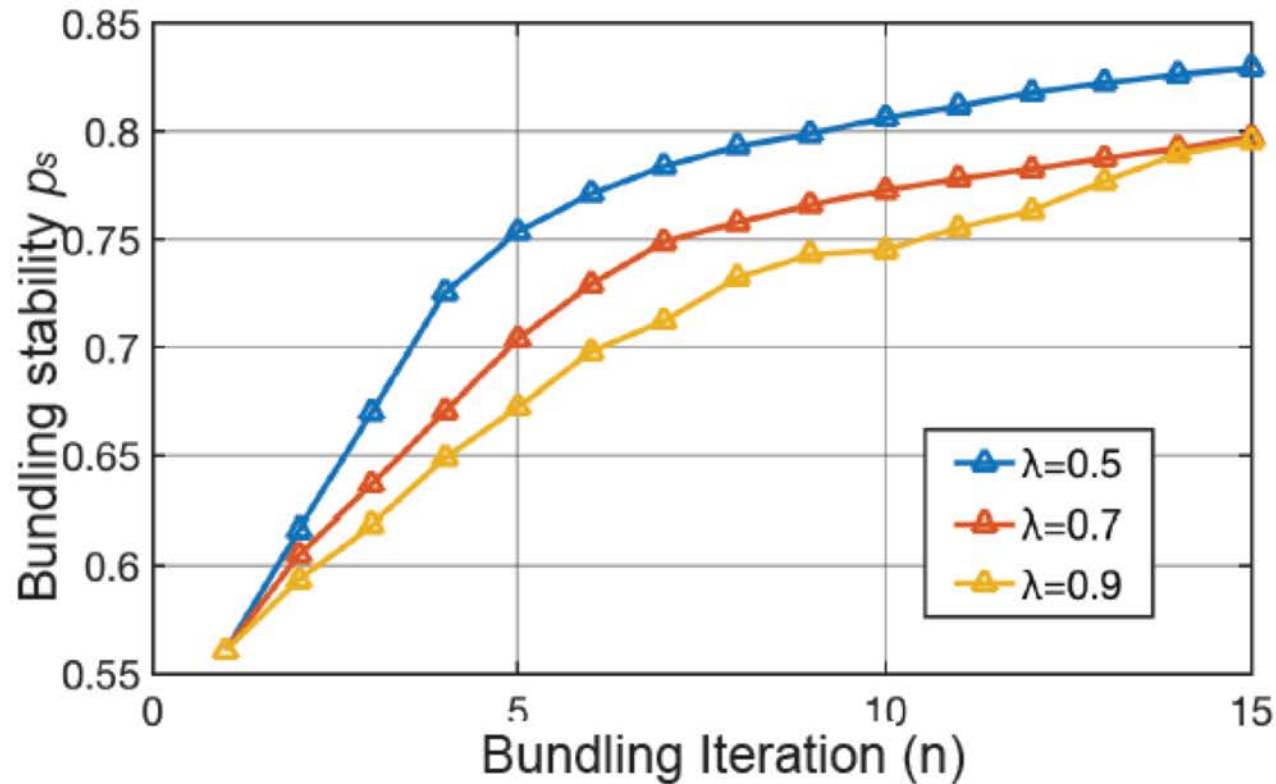
```
1: for  $i = 1$  to  $N$  do
2:   for  $j = i + 1$  to  $N$  do
3:      $d[i][j] = d[j][i] = \text{DiscreteFrechetDistance}(P_i, P_j)$ 
4:  $C = \text{DBSCAN}(P, \varepsilon, \text{minNum})$ ;
5:  $C_{max} = \text{argmax}_{C_i \in C} |C_i|$ ;
6:  $d_{geo} = 0$ ;
7: for each  $P_i \in C_{max}$  do
8:   for each  $P_j \in C_{max} \ \&\& \ i \neq j$  do
9:      $d_{geo} = d_{geo} + d[i][j]$ ;
10:  $p_r = d_{geo} / |C_{max}| / (|C_{max}| - 1) / 2$ ;
11: return  $p_r$ 
```



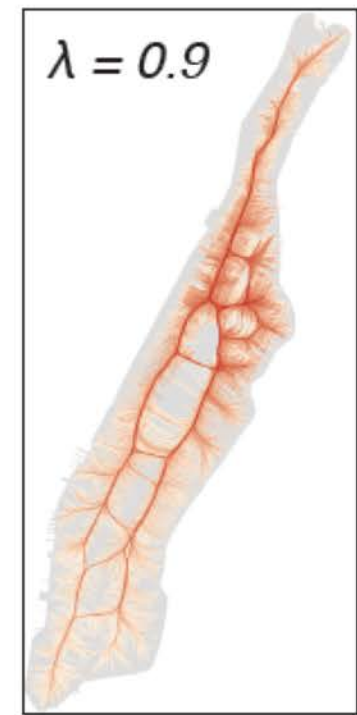
Evaluation

- Termination: Bundle stability (p_s) to determine when to stop iteration

- Bunc
$$NMI(X,Y) = \frac{2MI(X,Y)}{H(X) + H(Y)} \quad MI(X,Y) = \sum_{x \in X} \sum_{y \in Y} p(x,y) \log \left(\frac{p(x,y)}{p(x)p(y)} \right)$$



$n = 10$



$n = 11$

■ Introduction

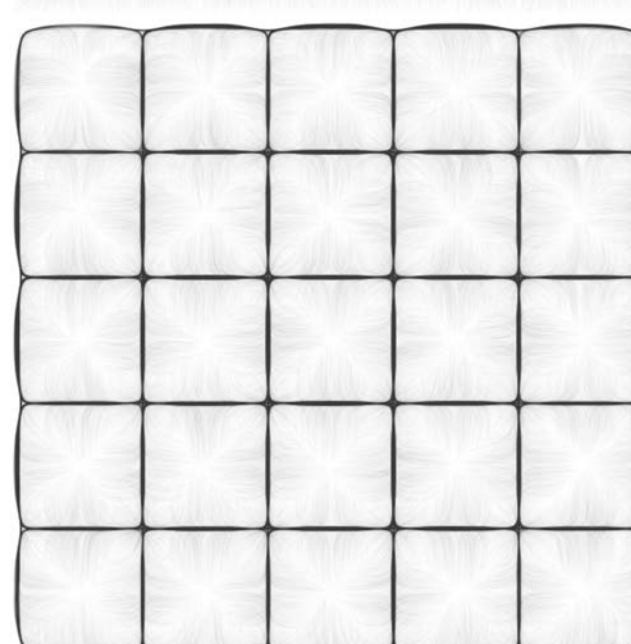
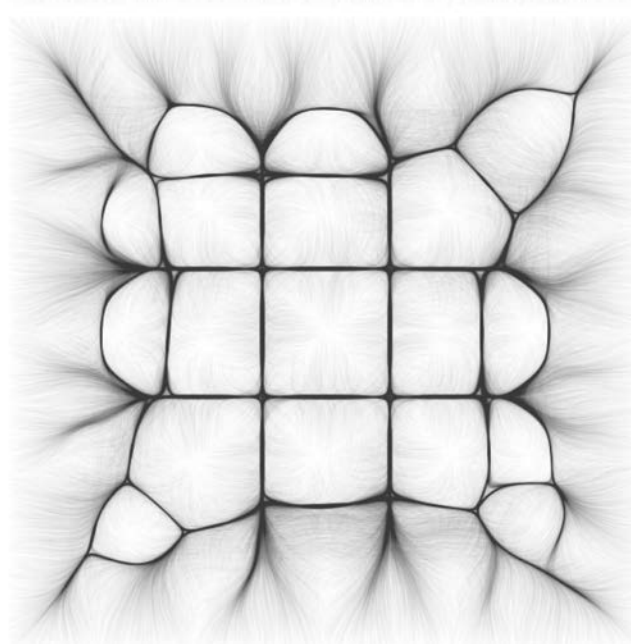
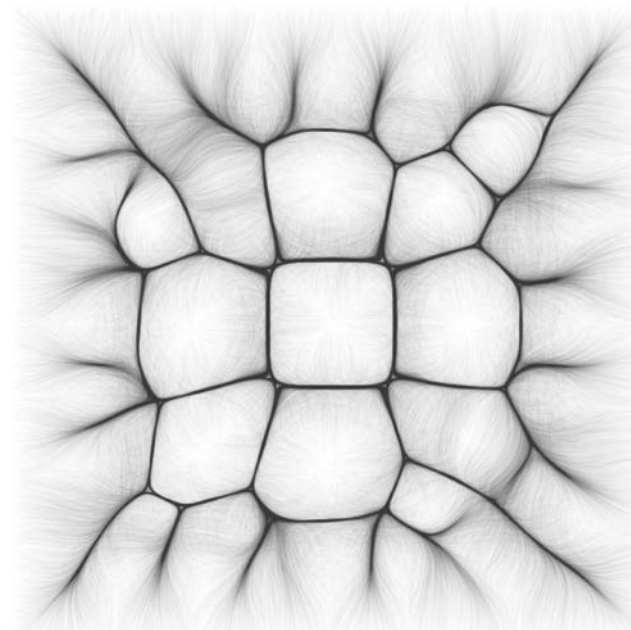
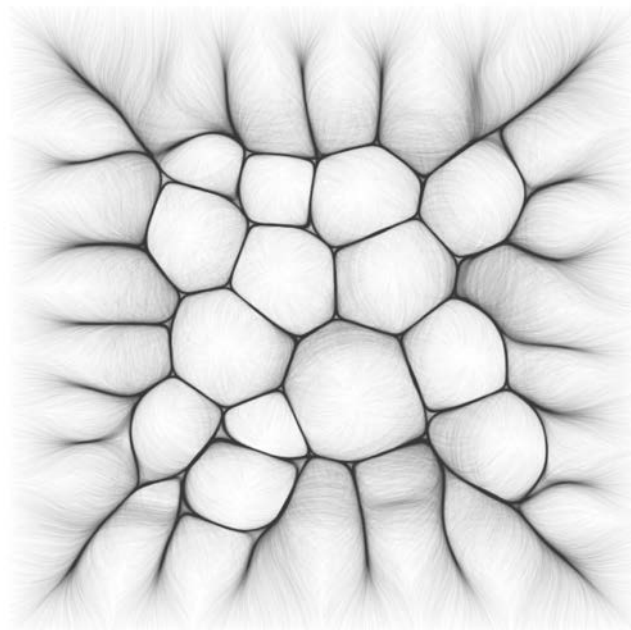
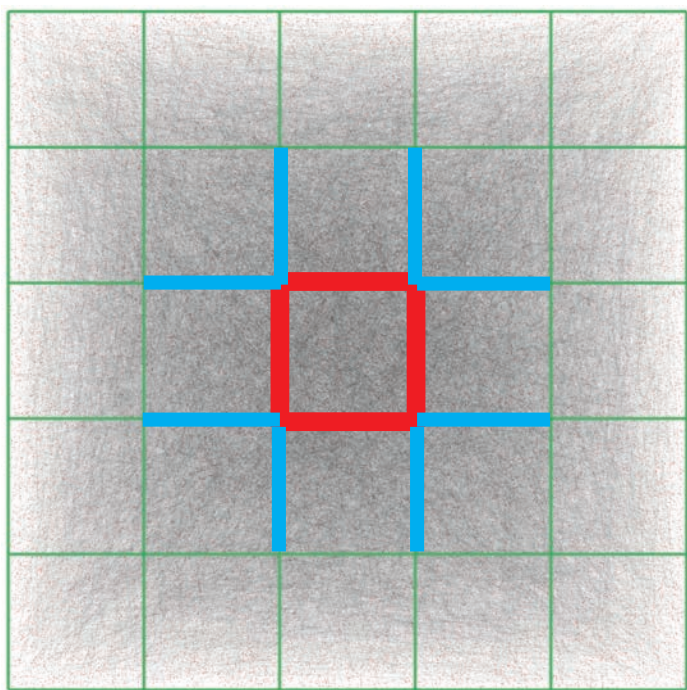
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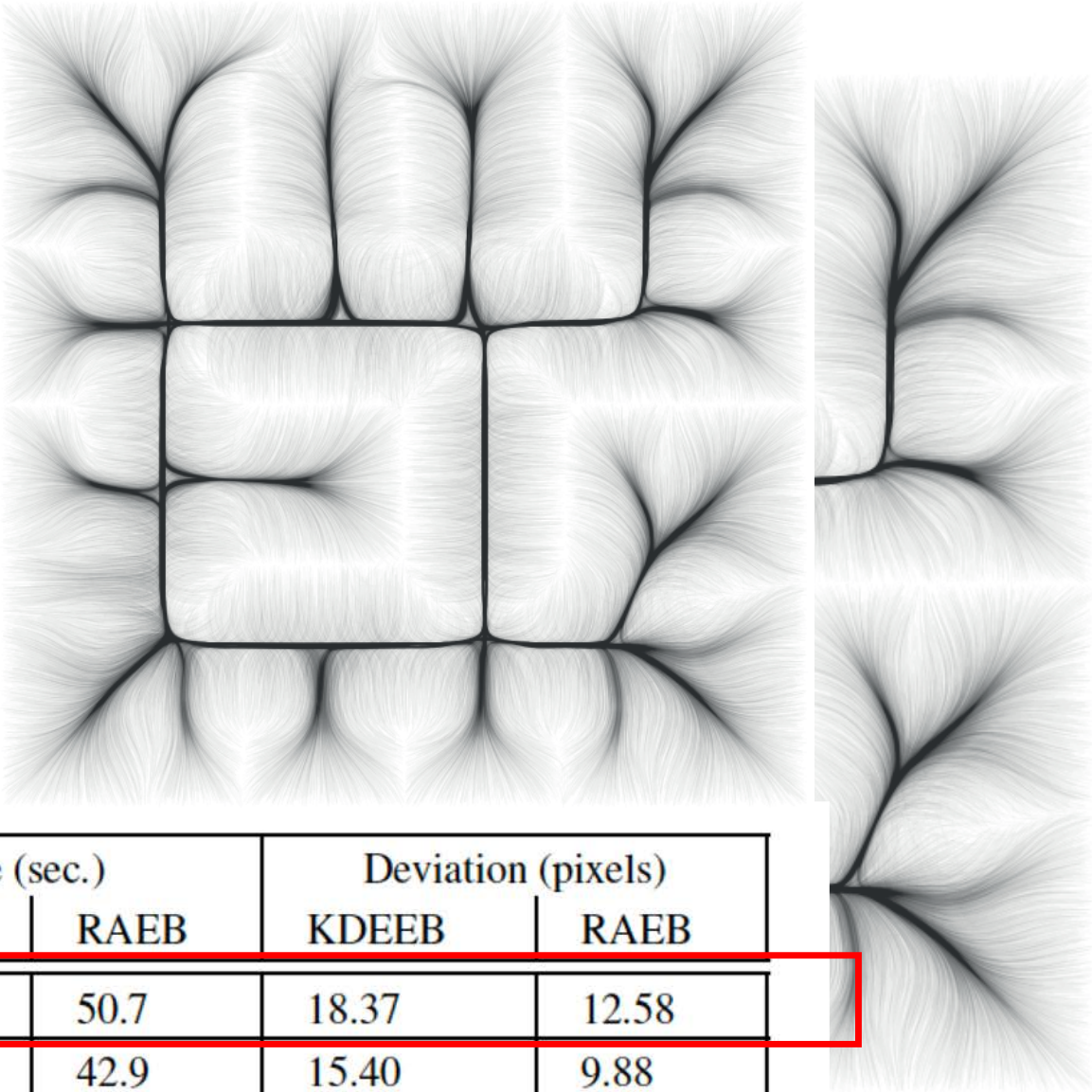
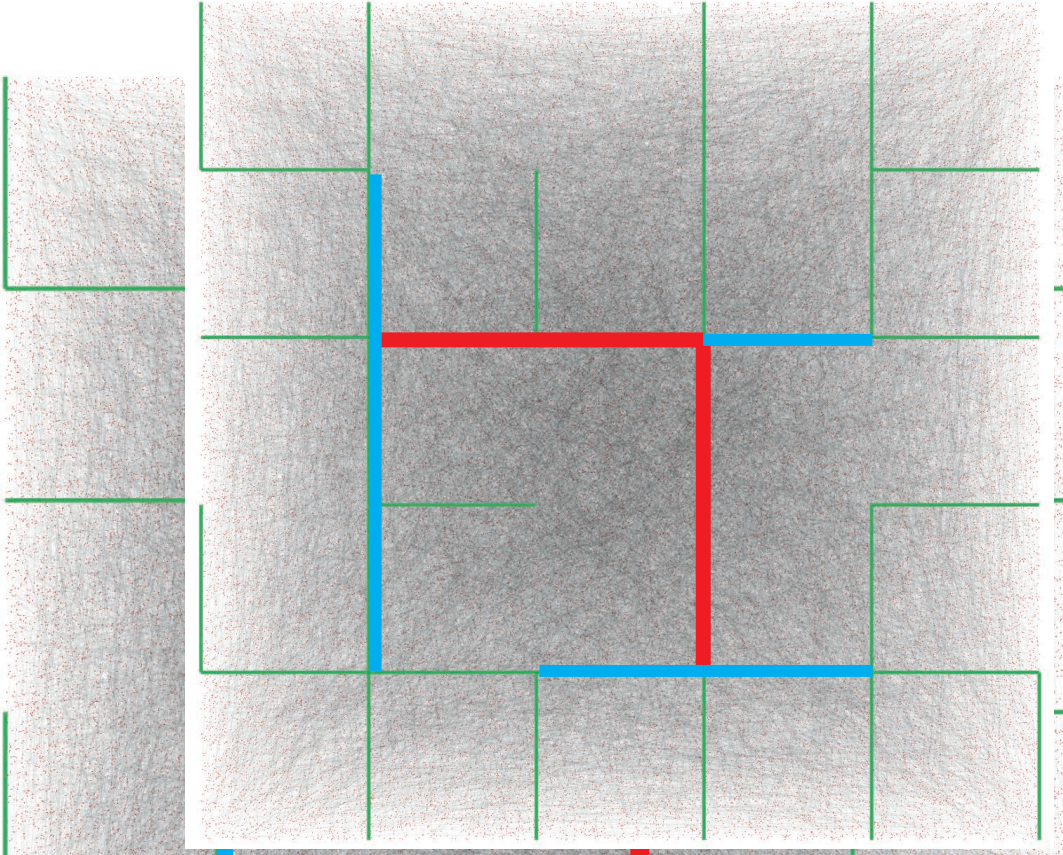
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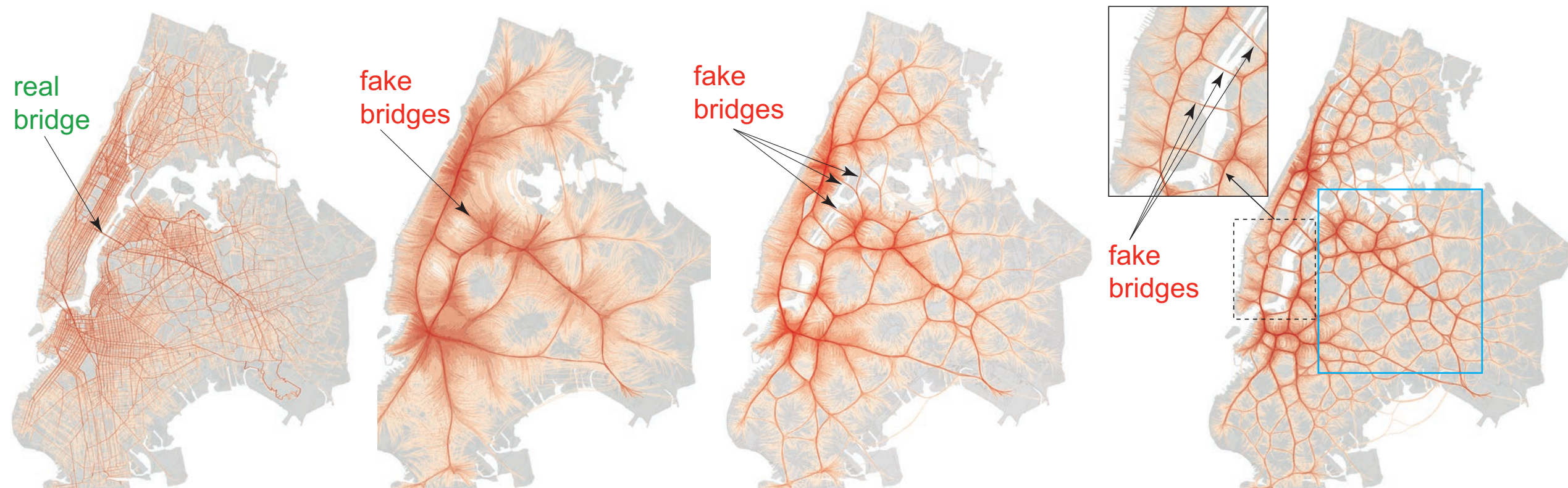


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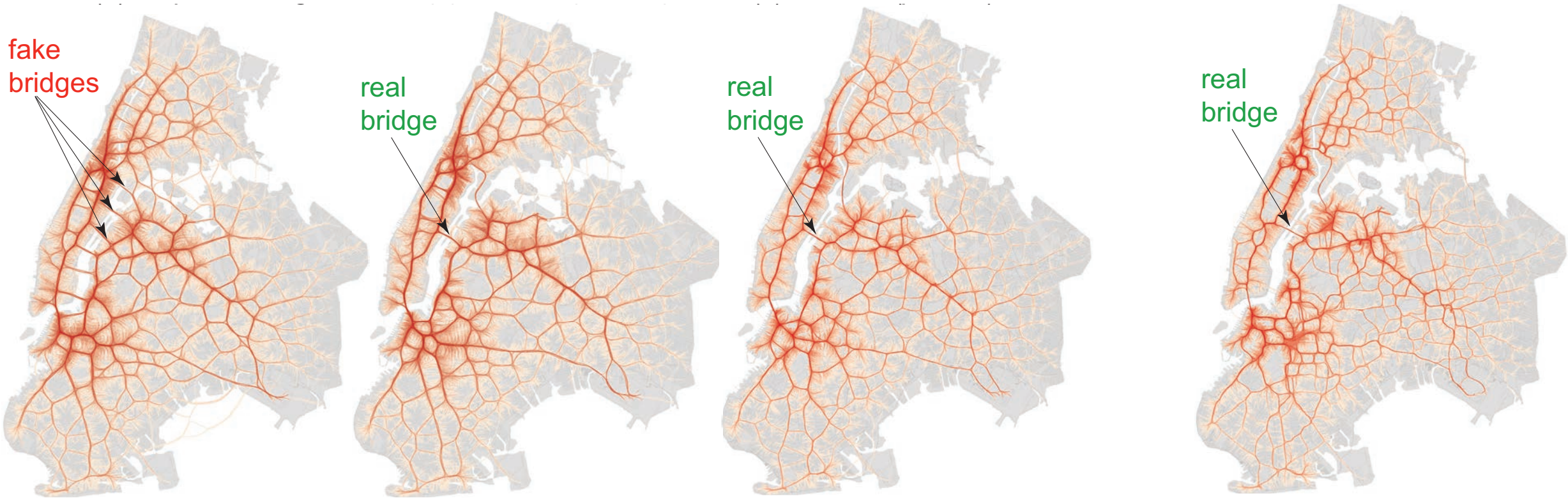


Dataset	Edge samples	p_n	Time (sec.)		Deviation (pixels)	
			KDEEB	RAEB	KDEEB	RAEB
Synthetic	4.6M	13	40.3	50.7	18.37	12.58
New York	3.1M	11	34.3	42.9	15.40	9.88
Shenzhen	1.3M	8	13.8	22.8	13.71	10.53

Application 2: NYC Taxi

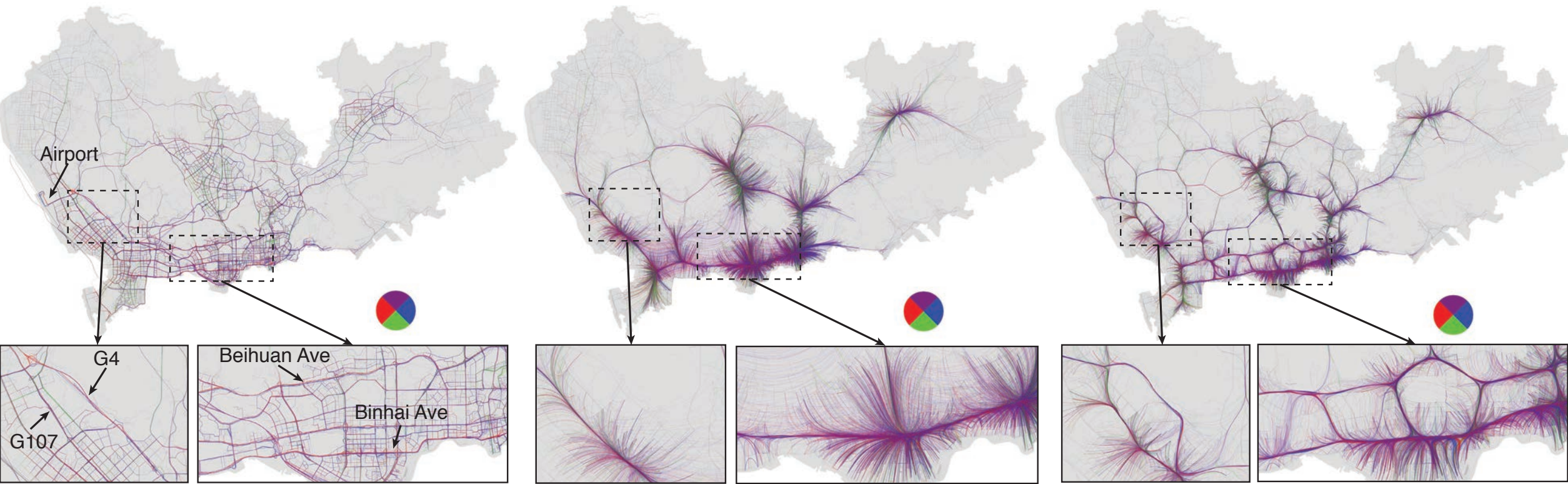


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Application 3: Shenzhen Taxi



(a) Map Matching

(b) KDEEB

(c) RAEB

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Discussions

- RAEB constrains trails to a given road network
 - **Route awareness (p_{ra})**: controls how bundles follow roads at a user-selected hierarchy level.
 - **Kernel size (p_r)**: determined by both the road network geometry and its resolution in image space.
 - **Bundling stability (p_s)**: automatically stops bundling when this similarity exceeds a given threshold.
- RAEB outperforms KDEEB on both synthetic and real OD trails
 - Visually more realistic
 - Quantitatively closer to ground-truth results
 - Comparable running time
- Limitations and future work
 - Visual hints on bundle deformation
 - Incorporate directional bundling techniques
 - Local and adaptive parameter settings: p_{ra} and p_r

谢谢！

Thank You!

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